

Vikhyat Soral

📍 Bengaluru, India ✉ vikhyatsoral2004@gmail.com ☎ +91-6376067811 in linkedin.com/in/vikhyat-soral-8a
🔗 github.com/Vikhyat03

Professional Summary

Motivated third-year B.Tech Electronics and Communication Engineering student with strong analytical and problem-solving skills and a passion for embedded systems, electronics, and intelligent engineering solutions. Experienced in signal processing, embedded hardware platforms, cybersecurity research, and electronic system analysis using MATLAB, Verilog and modern development tools. Eager to design, develop, and optimize reliable technologies that bridge hardware and software to solve real-world engineering challenges.

Education

Dayananda Sagar Academy of Technology & Management <i>Bachelor of Technology in Electronics and Communication Engineering</i> CGPA: 6.35 / 10	2023 – 2027 Bengaluru, Karnataka
Kendriya Vidyalaya No.1 AFS, Jodhpur (CBSE) <i>Senior Secondary (Class XII)</i>	2022 86.2%
Kendriya Vidyalaya No.1 AFS, Jodhpur (CBSE) <i>Secondary (Class X)</i>	2020 94.4%

Technical Expertise

Embedded Systems & Firmware Development

- Developed embedded firmware for **ARM Cortex-M3 (LPC1768)** microcontrollers using **Embedded C** in **Keil μ Vision**.
- Configured and interfaced **GPIO**, **UART**, and **I²C** peripherals for real-time communication and sensor integration.
- Designed a **Smart Temperature Logger** using an **I²C** temperature sensor with UART-based serial data acquisition and monitoring.
- Built a **Smart Home Automation** prototype using **FreeRTOS**, implementing multitasking, task scheduling, and GPIO-based device control.
- Programmed, tested, and debugged firmware on the **LPC1768** development board using **Flash Magic**.

VLSI Design & Digital IC Design

- Developed RTL designs using **Verilog HDL** in **Xilinx Vivado**, implementing finite state machines (FSMs), traffic light controllers, and fundamental digital logic circuits.
- Designed and simulated digital circuits in **Cadence Virtuoso**, including **32-bit ALUs**, counters, and flip-flops through schematic capture and waveform analysis.
- Designed a **CMOS inverter** at the transistor level and performed **DC** and **Transient** analyses to evaluate switching characteristics and circuit performance.

Programming Languages & HDL

- Embedded:** Embedded C, ARM Assembly
- Programming:** C, Python, MATLAB
- Hardware Description Language:** Verilog HDL

Development & Simulation Tools

- Keil μ Vision**, **Flash Magic**, **Xilinx Vivado**, **Cadence Virtuoso**, **MATLAB**, **Git**, **GitHub**, **VS Code**

Communication Protocols & Microcontroller Peripherals

- Worked with **UART** and **I²C** protocols for serial communication, debugging, and sensor interfacing.
- Configured and programmed **GPIO**, **ADC**, **DAC**, **Timers**, **External Interrupts (EINT3)**, and the **Nested Vectored Interrupt Controller (NVIC)** for peripheral control and interrupt-driven embedded applications.

Research Experience

AI-Based Integrated Multi-Layer Phishing Defense System with Automated Incident Response

Current Research

- Conducting a comprehensive literature review of state-of-the-art AI techniques for phishing detection across email, URL, visual content, and identity-based attacks.
- Analyzing deep learning architectures including **BERT**, **RoBERTa**, **CNN**, **LSTM**, **Vision Transformers (ViT)**, **Hybrid Deep Learning**, and **Multimodal AI** for phishing detection.
- Investigating **Explainable AI (LIME, SHAP)**, **Security Orchestration, Automation and Response (SOAR)**, and **Automated Incident Response (AIR)** frameworks for intelligent phishing detection and enterprise cybersecurity.
- Designing a unified multi-layer phishing defense architecture by integrating semantic analysis, URL intelligence, computer vision, Explainable AI, and automated remediation.
- Exploring research directions in adversarial robustness, privacy-preserving AI, encrypted communication security, and lightweight edge deployment for real-time phishing detection.

Projects

Hand Gesture Recognition using EMG Dataset | MATLAB

04/2025 – 06/2025

- Designed and developed an **Electromyography (EMG)**-based hand gesture recognition system for intelligent human-computer interaction.
- Performed signal preprocessing using digital filtering, rectification, normalization, and noise reduction techniques to improve signal quality.
- Extracted time-domain physiological features including **Mean Absolute Value (MAV)** and **Root Mean Square (RMS)** for accurate gesture representation.
- Trained and evaluated machine learning classifiers including **Support Vector Machine (SVM)** and **k-Nearest Neighbors (k-NN)** to classify multiple hand gestures.
- Improved classification reliability through feature engineering, signal analysis, and model validation using MATLAB.

Component Tolerance Modelling and Analysis | MATLAB

10/2025 – 01/2026

- Developed statistical models to analyze the influence of electronic component tolerances on analog circuit performance and manufacturing yield.
- Implemented **Monte Carlo simulations** to evaluate the effects of resistor and capacitor variations under multiple operating conditions.
- Applied **Extreme Value Analysis (EVA)** to identify worst-case scenarios and estimate circuit reliability.
- Performed probability-based performance evaluation to predict production yield and optimize component selection.
- Demonstrated the importance of tolerance-aware circuit design for improving robustness, reliability, and manufacturing efficiency.

Certifications

Generative and Agentic AI for Business Analytics

BITS School of Management (BITSoM), Centre of Executive and Professional Development *08/2025 – Present*

Internal Audit Job Simulation

Goldman Sachs (Forage)

07/2025

Awards & Achievements

Olympiad Gold Medalist (English & Mathematics)

Awarded by the Science Olympiad Foundation (SOF) for outstanding academic performance.

State Rank 4 – International Science Talent Examination

Recognized among more than 10,000 participants for exceptional scientific aptitude.

National Representative – 25th KVS Children's Science Congress

Represented the region at the national level by presenting an innovation project on Disaster Management.

Rajya Puraskar – Bharat Scouts and Guides

Completed advanced leadership, survival, teamwork and community service training.

Kalagaurav Award in Fine Arts

Recognized by the National Institute Kalabharti, Pune for excellence in painting and handwriting.